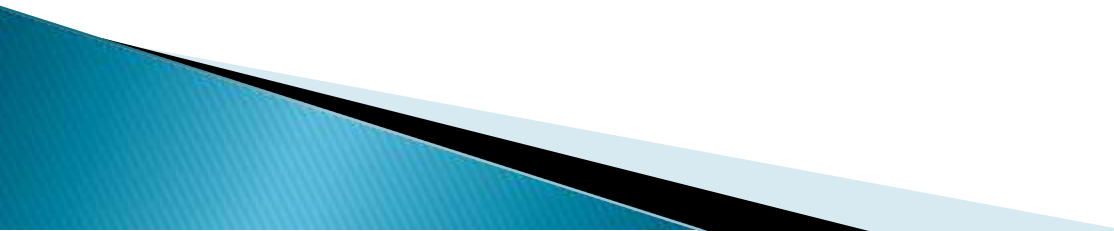


Topic: CONTROL KNOWLEDGE IN ARTIFICIAL INTELLIGENCE

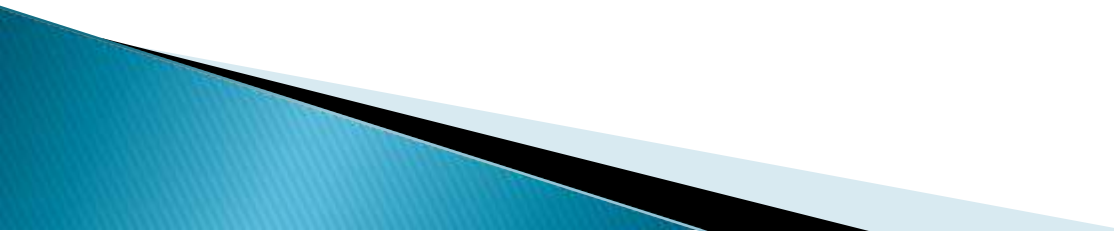
Presented By
DILSHADA FAROOQ
8th Semester
210324

**GOVERNMENT COLLEGE OF ENGINEERING AND
TECHNOLOGY SAFAPOORA GANDERBAL**

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- Introduction
 - What Control Knowledge does?
 - Why is it important?
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 - Real-Life Applications
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•Introduction

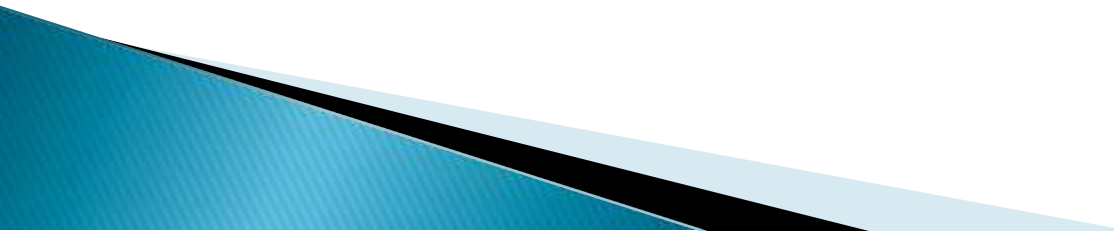
- ❖ Knowledge is the collection of facts, information, and relationships.
 - ❖ Without knowledge, AI is just raw computation
 - ❖ With knowledge, AI can understand context , solve problems and mimic human reasoning.
 - ❖ Knowledge(What the system knows)
 - ❖ Control knowledge(How it uses what it knows)
- 

•What Control Knowledge does?

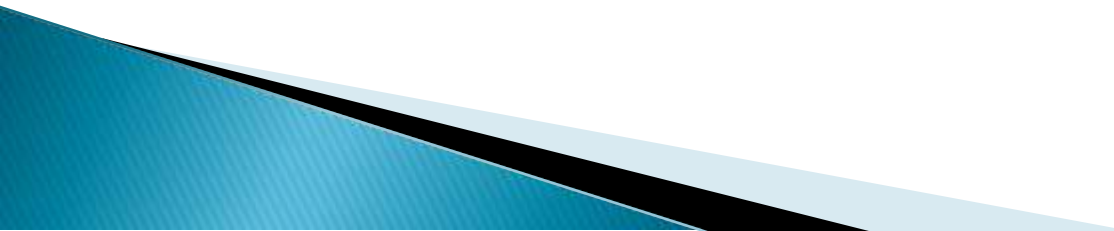
- ▶ Guides Decision-Making
- ▶ Manages Reasoning Processes
- ▶ Enables Adaptability
- ▶ Supports Autonomous Behavior
- ▶ Determines State Transitions



•Why is it important?

- ▶ Without control knowledge, an AI system may blindly search through every possible solution, which is inefficient .
 - ▶ Control knowledge makes AI systems behave more like humans—focusing only on the most promising paths .
 - ▶ Example: In chess, instead of calculating every possible move, control knowledge helps prioritize important strategies (like protecting the king).
- 

•Types :

1. **Knowledge about the Relevance of Information:** helps the AI system decide which information is important for the current problem and which can be ignored.
 2. **Knowledge about the Order of Rules:** determines the sequence in which rules or actions should be applied .
 3. **Knowledge about the Importance of Rules:** helps AI prioritize some rules over others based on their importance or impact.
 4. **Meta-Knowledge:** higher-level knowledge that helps AI decide which strategy or method to use for solving a problem
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•Real-Life Applications

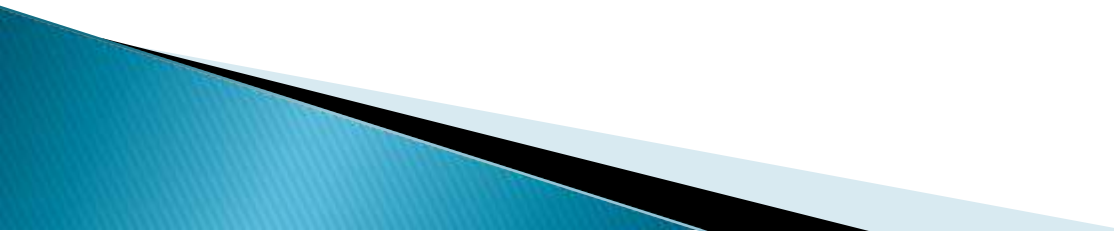
- ▶ Navigation Systems(Google Maps)
- ▶ Medical Diagnosis Systems.
- ▶ Robotics.
- ▶ Game AI

•Limitations

- ▶ Requires careful design and testing.
- ▶ Dependency on Correctness.
- ▶ Lack of Flexibility.



•Conclusion

- ▶ Knowledge is the foundation of intelligence in AI, and control knowledge is what makes problem-solving efficient and intelligent.
 - ▶ By guiding the order, relevance, importance, and strategy, control knowledge helps AI behave more like humans.
 - ▶ It reduces wasted effort, improves decision-making, and allows systems to adapt to different situations.
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Thank You

